

7.1. Proof of Theorems

This section provides the complete proofs of the theoretical results in Section 5, whose proofs were not included there.

7.1.1. Theorem 1.

Proof. We define $\tau^+(B, m)$ as the time elapsed until the first bug is detected in the m^{th} trial. It is worth noting that $\tau^+(B, 1), \tau^+(B, 2), \dots, \tau^+(B, M)$ are independently and identically distributed based on the definition, and thus the law of large numbers gives us

$$\lim_{M \rightarrow \infty} \frac{\sum_{m=1}^M \tau^+(B, m)}{M} = E[\tau^+(B, 1)] = E[\tau^+(B)]$$

and subsequently we have

$$\begin{aligned} \lim_{M \rightarrow \infty} \frac{\sum_{m=1}^M \tau^+(B, m)}{M} &= \frac{1}{P(X \in D : X \in B)} \\ &= \frac{1}{\pi(1)}. \end{aligned}$$

In the meantime, we have that

$$\begin{aligned} E[\tau^+(B)] &= E[\min \{t \geq 0 : X(t) \in B\}] \\ &= E[\min \{t \geq 0 : N_t = 1\}]. \end{aligned}$$

We further define

$$\tau^{+,1}(B) = \min \{t \geq \tau^+(B) : N_t = 2\}$$

and generally for any $n \geq 1$

$$\tau^{+,n}(B) = \min \{t \geq \tau^{+,n-1}(B) : N_t = n\}.$$

Based on the Markov property and the fact that the transition matrix is homogeneous in time, we derive that $\tau^+(B), \tau^{+,1}(B), \dots, \tau^{+,n}(B)$ are independently and identically distributed.

It is worth noting that N_T meets that

$$\sum_{n=1}^{N_T} \tau^{+,n}(B) < T \leq \sum_{n=1}^{N_T+1} \tau^{+,n}(B)$$

which implies that

$$\frac{\sum_{n=1}^{N_T} \tau^{+,n}(B)}{N_T} < \frac{T}{N_T} \leq \frac{\sum_{n=1}^{N_T+1} \tau^{+,n}(B)}{N_T}.$$

By leveraging the law of large number, we obtain that

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{\sum_{n=1}^{N_T} \tau^{+,n}(B)}{N_T} &= E[\tau^+(B)] \\ \lim_{T \rightarrow \infty} \frac{\sum_{n=1}^{N_T+1} \tau^{+,n}(B)}{N_T} &= E[\tau^+(B)]. \end{aligned}$$

Consequently, we have

$$\lim_{T \rightarrow \infty} \frac{T}{N_T} = E[\tau^+(B)],$$

or equivalently,

$$\lim_{T \rightarrow \infty} \frac{N_T}{T} = \frac{1}{E[\tau^+(B)]} = \Pr(X \in B | X \in D)$$

which completes last part of the statement, and thus concludes the proof. $\square$

7.1.2. Theorem 2.

Proof. This follows from [39]. $\square$

7.1.3. Theorem 3.

Proof. We note that

$$\begin{aligned} \tilde{P}_R(T) &= \frac{1}{M} \sum_{m=1}^M \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B} \end{aligned}$$

and if we take the expectation, we derive that

$$E[\tilde{P}_R(T)] = \frac{1}{T} \sum_{t=1}^T P_R^t(1).$$

Subsequently, we derive that

$$\begin{aligned} &|E[\tilde{P}_R(T)] - \pi(1)| \\ &= \left| \frac{1}{T} \sum_{t=1}^T P_R^t(1) - \pi(1) \right| \\ &\leq \frac{1}{T} \sum_{t=1}^T C_R \alpha_R^t \\ &\leq C_R \frac{1}{T} \frac{1}{1 - \alpha_R}. \end{aligned}$$

Subsequently, we characterize the difference between $E[P_R(x \in I_R^T : x \in H_R^T)]$ and $P_R(x \in I_R^T : x \in H_R^T)$. It is worth noting that $\frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B} \leq 1$ and we repeat the experiment for M times. Using Hoeffding's inequality, we immediately obtain that

$$P(|\tilde{P}_R(T) - E[\tilde{P}_R(T)]| \geq \sqrt{\frac{\log T}{T}}) \leq \frac{1}{T^2}.$$

This implies that with probability at least $1 - \frac{1}{T^2}$

$$\begin{aligned} & |\tilde{P}_R(T) - \pi(1)| \\ & \leq |\tilde{P}_R(T) - E[\tilde{P}_R(T)] + E[\tilde{P}_R(T)] - \pi(1)| \\ & \leq |\tilde{P}_R(T) - E[\tilde{P}_R(T)]| + |E[\tilde{P}_R(T)] - \pi(1)| \\ & \leq \sqrt{\frac{\log T}{T}} + \frac{1}{T} \frac{C_R}{1 - \alpha_R}. \end{aligned}$$

□

7.1.4. Theorem 4.

Proof. This follows from [40].

□

We also another alternative of analyzing nonreversible Markov chains. More specifically, we define $MP_R^X = \tilde{P}_R P_R^X$, where $\tilde{P}_R = \frac{P_R^X(x,y)\pi(x)}{\pi(y)}$ and let $\lambda_{M,R,2}$ and $\gamma_{M,R}$ denote the second-largest eigenvalue and the spectral gap of MP_R^X , respectively.

Theorem 13 (Spectral property for coupling MC (Poincare inequality)). *The result reads as based on [41] $\|(P_R^X)^T(x, \cdot) - \pi\|_{TV} \leq \frac{\lambda_{M,R,2}^T}{4\pi(x)} = \frac{(1-\gamma_{M,R})^T}{4\pi(x)}$.*

In other words, we recover the result in Theorem 2 with $C_R = \max_{\pi_0} \|\pi_0 - \pi\|_{L_2}$ and $\alpha_R = 1 - \gamma_R$.

7.1.5. Theorem 6.

Proof. The proof partly follows from that of Theorem 3. Specifically, we again have the following decomposition.

This implies that with probability at least $1 - \frac{1}{T^2}$

$$\begin{aligned} & |P_R(x \in I_R^T : x \in H_R^T) - \pi(1)| \\ & \leq |P_R(x \in I_R^T : x \in H_R^T) - E[P_R(x \in I_R^T : x \in H_R^T)] \\ & \quad + E[P_R(x \in I_R^T : x \in H_R^T)] - \pi(1)| \\ & \leq |P_R(x \in I_R^T : x \in H_R^T) - E[P_R(x \in I_R^T : x \in H_R^T)]| \\ & \quad + |E[P_R(x \in I_R^T : x \in H_R^T)] - \pi(1)|. \end{aligned}$$

For the first quantity, we again have the Hoeffding's inequality hold, which reads as

$$\begin{aligned} & P(|P_R(x \in I_R^T : x \in H_R^T) - E[P_R(x \in I_R^T : x \in H_R^T)]| \\ & \geq \sqrt{\frac{\log T}{T}}) \leq \frac{1}{T^2}. \end{aligned} \quad (1)$$

For the second quantity, we use the result from Theorem 1.2 in [34] by specifying the target function $g(X_t)$ as $g(X_t) = 1_{X_t \in B}$. This immediately gives us that

$$|E[P_R(x \in I_R^T : x \in H_R^T)] - \pi(1)| \leq \sqrt{\frac{4\delta_R}{T}}. \quad (2)$$

Combining (1) and (2) immediately leads to that with probability $1 - \frac{1}{T^2}$,

$$\begin{aligned} & |P_R(x \in I_R^T : x \in H_R^T) - \pi(1)| \\ & \leq |P_R(x \in I_R^T : x \in H_R^T) - E[P_R(x \in I_R^T : x \in H_R^T)] \\ & \quad + E[P_R(x \in I_R^T : x \in H_R^T)] - \pi(1)| \\ & \leq \sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}}, \end{aligned}$$

which concludes the proof.

□

7.1.6. Theorem 7.

Proof. The proof follows from that in [41].

□

7.1.7. Theorem 8.

Proof. The proof follows directly from the finite-time convergence result of $\tilde{P}_R(T)$. The first term ensures that $|\tilde{P}_R(T) - \pi(1)| \leq \epsilon$ with probability at least $1 - \frac{1}{T^2}$, and the second term guarantees that this probability is at least $1 - \alpha$. Consequently, we derive both terms and determine the minimal T_R that satisfies the condition for a specific fuzzer R , then select the maximum T_R across all fuzzers.

□

7.1.8. Theorem 10.

Proof. The proof follows from Theorem 8, with $D_R(T)$ replaced by $U_R(T)$.

□

7.1.9. Theorem 11.

Proof. The proof first leverage the result in [36] or [38]. By the theoretical guarantee established in these work, we have that with probability at least $1 - \beta$,

$$|\gamma_{dps}(R) - \gamma_R| \leq O\left(\frac{1}{\sqrt{T}}\right).$$

Subsequently, we derive that

$$\begin{aligned}
& |\hat{U}_R - U_R| \\
& \leq \left| \sqrt{\frac{\log T}{T}} + \sqrt{\frac{4}{T\gamma_{dps}(R)}} - \sqrt{\frac{\log T}{T}} + \sqrt{\frac{4}{T\gamma_R}} \right| \\
& \leq \left| \sqrt{\frac{4}{T} \left(\frac{\gamma_{dps}(R) - \gamma_R}{\gamma_{dps}(R) \cdot \gamma_R} \right)} \right| \\
& \leq \sqrt{\frac{4}{T} \left(\frac{O(\frac{1}{\sqrt{T}})}{\gamma_{dps}(R) \cdot \gamma_R} \right)} \\
& \leq O\left(\frac{1}{T^{\frac{3}{4}}}\right) \sqrt{\frac{1}{\gamma_{dps}(R) \cdot \gamma_R}}.
\end{aligned}$$

In other words, with probability $1 - \beta$,

$$\begin{aligned}
U_R & \leq \hat{U}_R + O\left(\frac{1}{T^{\frac{3}{4}}}\right) \sqrt{\frac{1}{\gamma_{dps}(R) \cdot \gamma_R}} \\
& \leq \sqrt{\frac{\log T}{T}} + \sqrt{\frac{4}{T\gamma_{dps}(R)}} + O\left(\frac{1}{T^{\frac{3}{4}}}\right) \sqrt{\frac{1}{\gamma_{dps}(R) \cdot \gamma_R}} \\
& \leq \sqrt{\frac{\log T}{T}} + \sqrt{\frac{4}{T\gamma_{dps}(R)}} + O\left(\frac{1}{T^{\frac{3}{4}}}\right) \sqrt{\frac{1}{\gamma_{dps}(R) \cdot \gamma_R}}.
\end{aligned}$$

Notably, the term $\sqrt{\frac{1}{T}}$ dominates $\frac{1}{T^{\frac{3}{4}}}$, allowing us to compare $\sqrt{\frac{4}{T\gamma_{dps}(R)}}$ across fuzzers for evaluation. This reduces the problem to comparing γ_{dps} or equivalently $\hat{U}_R$, thereby completing the proof.

□

7.1.10. Theorem 12.

Proof. The proof again follows from Theorem 8, again with $D_R(T)$ replaced by $U_R(T)$.

□